

Biological Sciences

Science - Systematic pursuit of knowledge

History of Science - empirical knowledge

Biology - Greek words “bios” and “logos”

- Study of Life

Branches of Biology

Anatomy – study of body parts and location

Physiology – study of functions of body parts

Morphology - forms

Histology - tissues

Biochemistry – study of chemical processes and transformations in living systems

Cytology – study of cells

Ecology – study of ecosystem

Genetics – study of genes and heredity

Microbiology – study of microorganisms

Mycology – study of fungi

Phycology – study of algae

Parasitology – study of parasites and their hosts

Taxonomy – Science of classification and naming organisms

Botany – study of plants

Zoology – study of animals

Ethology – animal behavior

Ichthyology – fishes

Ornithology – birds

Entomology – insects

Scientific Method:

1. Ask a question
2. State a hypothesis
3. Conduct an experiment
4. Analyze the results
5. Make a conclusion

Types of Variables:

Independent Variable – variable that is being manipulated (cause).

Dependent Variable – variable that changes depending on manipulations (effect or result).

Types of data:

- Nominal – normal counting, no ranking, tags and labels (Male=1, Female=2)
- Ordinal – categories that can be ranked (1st, 2nd, 3rd)
- Interval – no true meaning of zero (temperature)
- Ratio – has true zero (weight)

Robert Hooke- Discovered cell from a cork made from bark of an oak tree

- Coined the term cell (cellulae), meaning small room

Robert Brown- discovered the cell's nucleus

LOUIS PASTEUR (father of MICROBIOLOGY, PASTEURIZATION, Vaccines for Cholera, anthrax and rabies)
ALEXANDER FLEMING (Discoverer of PENICILLIN)
ROBERT KOCH (Germ Theory of Disease, mechanism of the disease TB, cholera and anthrax)
EDWARD JENNER (pioneer of SMALLPOX VACCINE)
ANTON VAN LEUWENHOEK (father of MICROSCOPY)

Cells

Two types of cells:

- Prokaryotic (Bacteria and Archaeobacteria)
- Eukaryotic (Protist, Fungi, Plant, Animal)

Prokaryotic- do not contain true nucleus

- DNA is located in a region called the **nucleoid**

Eukaryotic- contain true nucleus and membrane-bound nucleus

Somatic

- Refer to **all body cells** except sex cells
- Contains Diploid (2n) number of Chromosomes
- Undergo mitosis

Sex Cells/Gametes

- The sperm cell of the male and the egg cell of the female
- Contains haploid (n) number of chromosomes
- Undergo mitosis and meiosis

CELL THEORY

Schwann (1838), Scheiden (1839) and Virchow (1858)

1. **All organisms are composed of CELL/S.**
2. **The CELL is the basic unit of structure and function.**
3. **CELLS arises from pre-existing cells.**

Cellular Parts

1. Plasma Membrane

- selective barrier (semi-permeable)
- control the passage of nutrients and waste
- boundary between the cell and its external environment

2. Nucleus

- Genetic library of the cell
- Contains most of the genes (DNA- Deoxyribonucleic acid)
- Largest organelle in the animal cell

3. Cytoplasm

- jelly-like fluid
- where organelles are located
- Compose of water, salts, proteins, fats, carbohydrates, and other chemicals

4. Ribosomes

- Protein factories in the cell
- Carry out protein synthesis

5. Endoplasmic Reticulum

- biosynthetic factory
- Rough ER (with ribosomes attached), Smooth ER (without ribosomes attached)
- Folded membranes that serves as **channels** through which materials are transported in the cell

6. Golgi Apparatus

- Shipping and receiving center
- Packaging factories of the cell

7. Mitochondrion

- Chemical energy conversion
- powerhouse of the cell
- Converts food molecules into energy (ATP) through **cellular respiration**
- ATP- Adenosine Triphosphate: Energy currency of the cell.
- The **waste products of cellular respiration are water and carbon dioxide gas**

8. Lysosomes

- Digestive compartments: garbage collector
- membranous sac of strong hydrolytic enzymes
- Digest bacteria, other foreign substances, and worn- out cells

9. Chloroplast

- Capture light energy
- Found in plants **NOT** in animals
- Site of photosynthesis

***Photosynthesis** – process of capturing light energy to produce food- energy rich organic molecules (plants, algae, and other chlorophyll-bearing organisms) from carbon dioxide.

Raw materials: sunlight, water, carbon dioxide

Waste product: Oxygen

End product: Sugar (glucose)

10. Cell Wall

- Protective layer of the cell
- Plant cell wall made up of **cellulose** (most abundant organic material on Earth)
- Fungal cell wall made up of **chitin**

11. Central Vacuoles

- found in plants
- hold reserves of important organic compounds and liquid
- take up to 30%-90% of cell's volume

12. Cytoskeleton

- mechanical support, network of fibers
- Microfilaments** – **thin structures** essential for cytokinesis, amoeboid movement and changes in cell shape

13. Cilia

- hair-like structure**
- example: cilia in paramecium, fallopian tube (oviduct), lungs

14. Flagella

- whip-like structure**
- example: flagellum in sperm cell, euglena, bacteria

Carbon, Hydrogen, Oxygen, Nitrogen- constitute 98% of living matter

Cell Division

Meiosis

- reductional
- four haploid daughter cells
- production of gametes/sex cells
- genetic diversity due to crossing over

-Meiosis 1 & 2 (Prophase I, Metaphase I, Anaphase I, Telophase I, Prophase II, Metaphase II, Anaphase II, Telophase II)

Mitosis

- vegetative
- two diploid daughter cells
- growth and repair of somatic cells
- Prophase, Metaphase (Middle), Anaphase (Separation), Telophase**
- Genetically identical with the parents

ANEUPLOIDY

Down's Syndrome (trisomy 21)

Edward's Syndrome (trisomy 18)

Patau's Syndrome (trisomy 13)

Klinefelter's (XXY)

Turner's (XO)

Cellular Transport

1. Passive Transport

- Diffusion of substance across a membrane with **no energy investment**
- Move from **high to low concentration (concentration gradient)**

Examples:

- Diffusion – tendency for molecules of any substance to **spread out evenly** into the available space.
- Facilitated Diffusion – **aided by transport proteins** (speed the movement of molecules across the plasma membrane)
- Osmosis - movement of water across a semipermeable membrane (**diffusion of water**)
 - affected by the concentration gradient of dissolved substance (**solution's tonicity**).
1. Hypotonic solution - concentration of solute is **lower** than inside the cell
 - water comes inside the cell
 - **clues: turgid, swell, burst, lysed** (Hypotonic- *pumutok si* cell)
2. Isotonic solution - concentration of solute outside is the **same** with inside the cell
 - **NO NET movement of water**
3. Hypertonic solution - concentration of solute outside is **greater** than inside the cell.
 - water goes outside the cell
 - **clues: plasmolysis, shrink, shrivel, dehydrated**

2. Active Transport

- Uses energy** to move solutes against their concentration gradients
- Requires energy**, usually in the form of **ATP**
- Examples:** Ion pumps, cotransport, bulk transport (exocytosis, endocytosis)

3. Bulk Transport

- transport of large molecules
- phagocytosis (cell eating)
- pinocytosis (cell drinking)

Common elements in living things:

- Oxygen (O) 65% of Human body weight
- Carbon (C) 18%
- Hydrogen (H) 10 %

- Nitrogen (N) 3%
- Hydrogen – most abundant element in the universe
- Nitrogen – most abundant gas in the atmosphere
- Helium – 2nd most abundant element in the universe
- Oxygen – 65% of Human body weight, Carbon (C) 18%
- Hydrogen (H) 10 %
- Nitrogen (N) 3%
- Calcium - most abundant mineral in the human body (bones)

Functional Groups:

HYDROXYL GROUP= alcohols

CARBONYL GROUP= ketones

CARBOXYL GROUP= carboxylic acid, acetic acid

AMINO GROUP= proteins

THIOL GROUP= sulfides

PHOSPHATE GROUP (PO₄⁻)

PHENYL GROUP= benzenes, aromatic hydrocarbons

Macromolecules

1. Carbohydrates

- Building blocks are monosaccharides (glucose, fructose)
- Main source of Energy in the body

Monosaccharides (simple sugars)

Glucose/dextrose (blood sugar)

Fructose (fruit sugar)

Galactose (milk sugar)

Disaccharides

Sucrose (table sugar, cane sugar): Glucose + Fructose

Lactulose: Galactose + Fructose

Lactose (milk sugar): Galactose + Glucose

Maltose (malt sugar): Glucose + Glucose

Polysaccharides

- energy storage (animal glycogen and plant starch)
- structure support (plant cellulose).
- Starch, glycogen, cellulose

2. Nucleic Acids

- Building blocks are nucleotides
- 3 basic parts:
 - Nitrogenous base (Purines and Pyrimidines)
 - Phosphate group
 - Sugar

****Deoxyribonucleic Acid (DNA)**

- double helix
- contains the biological information
- Adenine- Thymine, Cytosine-Guanine pairing

****Ribonucleic Acid (RNA)**

Your license awaits you. Dream big. Pray hard. Put your best foot forward. Trust yourself. Leave the rest to Him.

- single stranded
- Adenine- Uracil, Cytosine-Guanine

Chargaff rule- number of adenine is equal to thymine, likewise with cytosine and guanine.

Central dogma of life:

1. Transcription- creation of mRNA out from a template DNA
2. Translation- translating mRNA to amino acids forming proteins

Mutation

- is a change in gene or chromosomes, which caused a new trait to be inherited

3. Proteins

- Building blocks are amino acids
- joined by peptide bond
-

4. Lipids - Building blocks are glycerol and fatty acids

Taxonomy

8 Levels of Taxonomic Classification

Dream	Domain
Ko	Kingdom
Pumasa	Phylum
Coz	Class
Our	Order
Family	Family
Gets	Genus
Sweldo	Species

Carolus Linnaeus

- Father of modern Taxonomy
- Proposed the binomial nomenclature

2 Kingdom System

- Aristotle
- Plantae and Animalia

3 Kingdom System

- Ernst Haeckel
- added Protista

4 Kingdom System

- H.F Copeland
- added Monera

5 Kingdom System

- Whittaker
- added Fungi

Kingdoms and Domains

Virus - not living, inert chemicals

- has **capsid**- protein shells that protects the virus' DNA or RNA.

Your license awaits you. Dream big. Pray hard. Put your best foot forward. Trust yourself. Leave the rest to Him.

Types of Viruses

1. Bacteriophages invade only bacterial cells

2. DNA Virus

- Contains DNA as the genetic materials
- Ex. Viruses that cause sore throat, warts, herpes, chicken pox and small pox

3. RNA Virus

- Contains RNA as the genetic material
- Ex. Viruses that cause polio, common colds, hepatitis, and rabies

4. Retrovirus

- Contains an enzyme, **reverse transcriptase** that makes DNA from RNA
- Ex. HIV

1. Domain Archaea:

a. Kingdom Archaeobacteria

- live in various places, some in **severe environments**
- examples: methanogens, halophiles, thermopiles

2. Domain Bacteria:

a. Kingdom Eubacteria (Monera)

- referred to as the true bacteria and usually called the **“bacteria”** group
- example: streptococcus

****Bacterial Morphology**

Cocci- circle, round

Bacilli- rod- shaped

Vibrio- comma- shaped

Sprochete- spiral

3. Domain Eukarya

a. Kingdom Protista

- protozoans, diatoms, various types of algae (green, golden, brown, red algae)
- examples: entamoeba histolytica (amoebiasis)

b. Kingdom Fungi

- are **saprophytic** (decomposers)
- fungi, molds, mushroom, yeast, microsporidia

c. Kingdom Animalia

- major groups: invertebrate and vertebrate
- all are heterotrophic or consumers

c.1. Porifera (Phylum Porifera) - pore-bearing, simplest animal, body of calcium carbonate. Examples: sponges

c.2. Cnidaria (Phylum Cnidaria) - **has stinging cells (nematocysts)** Examples: jellyfish, corals, hydra

c.3. Arthropoda (Phylum Arthropoda) - **jointed legs**, segmented bodies. Examples: Insects (3 pairs of legs) - mosquito, butterflies; Arachnids (4 pairs of legs) - spiders; Crustaceans (5 pairs of legs) - crabs, shrimps, lobsters; Centipede; Millipede

***Molting or ecdysis - insects shed off their old cuticle to grow in size**

c.4. Nematoda (Phylum Nematoda)- **roundworms, unsegmented**. Examples: pinworm, ascaris, hookworm, filarial worm

c.5. Platyhelminthes (Phylum Platyhelminthes) - **flatworms**: worms have soft tissues; unsegmented. Examples: flukes, tapeworms, planaria

c.6. Annelida (Phylum Annelida) - **segmented worms**. Examples: **leeches (anticoagulants), earthworms (hermaphrodite)**

c.7. Mollusca (Phylum Mollusca) - **soft-bodied animals (some are with shells)**. Examples: squids, clams, octopus

c.8. Echinodermata (Phylum Echinodermata) - spiny body. Examples: sea star, sea urchin, sand dollars, sea cucumber, brittle star

c.9. Chordata (Phylum Chordata) -
fish (pisces) - scales, gills and fins
amphibians - part of their life cycle is in water
reptiles - with scaly bodies
birds (aves) - 2 scaly legs, wings, feather
mammals - mammary glands, hairs, milk

***Walrus** - marine mammals that have a long ivory tusks and lives in arctic environment

Animal Tissues

1. Epithelial

- Covering of external surface and lining of internal surface

2. Connective (tendons, ligaments, cartilage, adipose, bone, blood)

- Binds structure together
- Provides support and protection

3. Muscle (skeletal, smooth, cardiac)

- For contraction to produce movement

4. Nervous Tissue

- For reception of stimuli and transmission of impulses
- Supporting cells for nourishment, protection and insulation of the neurons

3. Kingdom Plantae

-Autotrophic or producer

-consists of vascular (with tissues for transport of food and water) and non-vascular (without)

Nonvascular Plants

-No vascular or conducting tissues

-Found in moist places

-Examples: mosses, liverworts, hornworts

Vascular Plants

a. Seedless vascular plants (includes **ferns**, club mosses, spike mosses, quillworts, horsetails)

b. Seed vascular plants

b.1. Angiosperms (**flowering plants**; most abundant and distributed plant; bears fruit to protect the seed, flowers contain **reproductive** cells)

-monocot (**fibrous root, one cotyledon, parallel veins, flowers are in multiple of 3, shrubs**)

-dicot (**taproot system, 2 cotyledons, netted venation, flowers are in multiple of 4 or 5, woody**)

b.2. Gymnosperms (**cone-bearing; have naked seeds, seeds are not enclosed by a fruit**).

Examples: **conifers, cycads, ginkgoes, and gnetophytes**)

Vascular plants have **xylem (conducts water and mineral)** and **phloem (distribute sugars, amino acids and other organic compounds)**

Plant Tissues

DERMAL

-protection against water loss, regulates gas exchange, secretes metabolic compounds, and absorbs water and mineral nutrients

VASCULAR

- transport fluid and nutrients internally
- xylem and phloem

GROUND

- Parenchyma
 - Form the mesophyll in leaves, responsible for photosynthesis and the exchange of gases
 - Storage of starch, protein, fats, oils and water in roots, tubers, seed endosperm and cotyledons
 - Secretion
 - Wound repair
- Collenchyma
 - provide structural support
- Sclerenchyma
 - principal supporting cells in plant tissues that have ceased elongation
 - constitute the source material for many fabrics (fibers)

Tropism

- Plant growth responses

- Positive (+) towards the stimulus,
- Negative (-) away from stimulus

Types:

- Phototropism – growth response to light
- Geotropism –response to gravity
- Thigmotropism –response to touch
- Chemotropism –to chemical stimulus
- Hydrotropism –to water or soil moisture
- Cryotropism – growth response to cold

Plant Hormones

Growth Promoters

Auxin

- For cell elongation
- Produced in the shoot apical meristem

Giberillins

- For stem elongation
- Can cause dwarf plants to grow, seeds to germinate

Cytokinins

- Promote cell division
- Prevent death of leaves and senescence, initiates leaf growth

Growth Inhibitors

Abscisic acid

- Leaf abscission, also called stress hormone
- Initiates and maintain seed dormancy

- Brings about closing stomates when plant is under water stress

Ethylene

- For fruit ripening
- Involved in abscission (dropping of leaves of fruits and flowers)

Flowers- reproductive organ of flowering plant

Pollination- transfer of pollen from anther to the stigma

*cross pollination- one plant to the other

*self-pollination- same flower

Parts of a flower

-Sepals- protect the flower bud before it opens, collectively called **calyx**

-Petals- attract pollinators, collectively called **corolla**

-Stamen- **male** part

*anther- saclike container

*filament- a slender stalk

*pollen grains- develop from microspores developed in the anther

-Carpel (pistil)- female part

*stigma, style, ovary (becomes the fruit), ovule (becomes the seed), fruit (for seed distribution)

Parts of a Seed

Seed coat (Testa)

-Protective structures that reduce evaporation of water

-Protects from attack of fungi and insects

Embryo

-Miniature plant of the seed

Endosperm

-Stores food, starch proteins and oils

Asexual Reproduction - vegetative reproduction

Runners- **horizontal stem**, also known as **stolon**. Examples: strawberry, Bermuda grass, **bamboo grass**

Tubers- **underground stem** that stores food for the plant, the **"eyes" are the stem nodes**. Example: **potato**

Rhizomes- **horizontal underground stem**. Example: **ginger**

Bulbs- shortened, compressed **underground stem** surrounded by fleshy scales (leaves). Examples: **onion, garlic**

Corms- **underground stem**, shaped like bulbs but do not contain fleshy scales, **solid, swollen stem with dry, scale-like leaves**. Example: **taro(gabi)**

Underground roots- carrots, sweet potatoes, and cassava

Earthworm

- hermaphroditic- both male and female organs are found

- clitellum- contains egg

- **one worm passes its sperms into the seminal vesicles of another, at the same time receives sperms into its own receptacle**

Frog

-eggs are fertilized in water

-does not require a hard shell to protect its eggs

Budding- forming outgrowth called bud. Examples: hydra, sponges, yeast

Fragmentation- separated pieces of parent organism develop into an individual. Examples: sea star, flatworms

Binary fission- in bacteria

Human Systems

1. Digestive System

- Mouth (**chemical and mechanical digestion**)
- pharynx- throat
- esophagus- **peristalsis** occur (wave-like muscle contraction that moves food)
- stomach- **chyme** (mixed food and gastric juices)
- small intestine- **final place of digestion**
- large intestine- colon, water absorption, production of vit K, ends in rectum
- anus- defecation
- accessory organs:
 - salivary gland
 - liver- largest internal organ in the body (emulsification of fats)
 - gall bladder- stores bile
 - pancreas- secrete pancreatic amylase for final digestion of starch

2. Circulatory System

- Transport blood from heart to the lungs to oxygenate blood then to other parts of the body
- Vein- transports blood **B**ack to the heart
- Artery- transports blood **A**way from the heart
- Heart- pumps blood
- Blood- RBC, WBC, Platelets, Plasma

****RBC (red corpuscles) red because of hemoglobin**

****fibrinogen- protein for blood clotting**

****vitamin K- vitamin for blood clotting**

****calcium- mineral for blood clotting**

****Vitamin B12- vitamin for RBC production**

3. Respiratory System

- supplies body with oxygen and release carbon dioxide as waste

****inspiration- inhales, takes in oxygen**

****expiration- exhales, release carbon dioxide**

- nose- warms and moistens the air, traps dirt
- pharynx- passageway of both air and food
- epiglottis- prevents the entrance of food to trachea
- larynx- voice box
 - vocal cords are found inside the larynx (voice box)
- trachea- windpipe
- lungs- main organ for respiration
- diaphragm- dome-shaped muscle beneath the lungs
 - *inhales- contracts
 - *exhales- relaxes

4. Endocrine System

- Regulates** body temperature, metabolism, development, homeostasis through **hormones**.
- Example: Pancreas release insulin

5. Reproductive System

- Fertilization- union of sex cells (gametes)
- in vitro fertilization- sperm and ovum meet outside the body, in a test tube
- in vivo fertilization- sperm and ovum meet inside the body

6. Nervous System

-regulates behavior, maintains homeostasis, **controls sensory and motor functions**

-memory and emotions

-Neurons- the basic unit of nervous system

*Central Nervous System (Brain and spinal cord)

-cerebrum- voluntary activities, largest, intelligence, learning adjustment

-cerebellum- involuntary, balance and coordination

-brain stem- connects spinal cord and brain, involuntary, life-sustaining activities:

breathing, heart rate, blood pressure, and swallowing; midbrain, pons, medulla oblongata

*Peripheral Nervous System

- Somatic NS
 - Peripheral nerves (motor and sensory nerves)
- Autonomic NS (involuntary function)
 - Sympathetic (fight or flight)
 - Parasympathetic (rest and digest)

7. Skeletal System

-Protects and supports body parts

-framework

-blood cell production on bone marrows

-calcium and phosphorous storage

-bones (206 in adult)

8. Muscular System

-Moves limb and trunk

-Moves substances through the body

-600 muscles

****Types:**

-Smooth- involuntary, non-striated (example: intestine)

-Cardiac- involuntary, striated (example: heart)

-skeletal- voluntary, striated (muscles attached to bones for movement)

9. Excretory System

-Removes metabolic waste from the body

-kidney- main excretory organ; produces urine

-lungs- releases water vapor and carbon dioxide

-liver- produces urea

-skin- produces sweat

10. Integumentary System

-body covering

-protects from loss of internal fluids and from UV rays

-body temperature regulation

-nail, skin, hair

****Skin**- the largest organ in the human body

Death: Postmortem Changes

1. Livor mortis

- known as hypostatis
- blood settles because of gravity
- skin becomes **pinkish, bluish, botchy**

2. Rigor Mortis

- muscle mass becomes turgid**

3. Algor Mortis

- body cooling**

Decomposition- process of autolysis and putrefaction

Autolysis- self-digestion, destruction of cells through the action of enzymes

Putrefaction- breakdown of tissues and liquefaction of organs

Biotechnology

-Modification of biological process through human intervention

-Genetic engineering

-the recipient becomes GMO (genetically modified organism) - transgenic organism

-Bt corn (genetically modified maize with toxin against corn borer, uses the genes of

Bacillus thuringensis.

-Bioremediation- use of biotechnology to solve environmental problems

Example: **oil eating bacteria- in cleaning up oil spills**

Ecology

Autotrophs- Producers, can do photosynthesis e.g plants

Heterotrophs- Consumer e.g humans

Photosynthesis

- a redox reaction
- harnessing energy from the sun to create glucose/sugar
- oxygen as a waste product

Cellular respiration

- a combustion reaction
- converts glucose to ATP
- needs oxygen, carbon dioxide as a waste product
- 1 molecule of glucose 38 ATPs (Glycolysis- 2 net ATPs, Krebs Cycle- 2 ATPs, ETC- 32 ATPs)

Levels of Biological Organization

Biosphere

Biome

Ecosystem

Community

Population

Organism

Organ System

Organs

Tissues

Cells

Organelles
Molecules
Atoms

Components of Ecosystem:

Components:

- BIOTIC (Living)
- ABIOTIC (Non-living)
- Transfer of energy

Place: Habitat

Role: Niche

Species Interaction

- MUTUALISM (+/+)
- PARASITISM (+/-)
- COMMENSALISM (+/0)
- COMPETITION (-/-)
- AMMENSALISM (-/0)
- PREDATION (w/ act of killing)
- GRAZING (only a part is eaten)

Biome:

- A climatically and geographically contiguous area with similar climatic conditions

-Polar

- Arctic (north), Antarctic (south)
- animals: polar bears, penguins, walrus

-Tundra

-permafrost

-Taiga

-Coniferous forest dominated by cone-bearing trees such as spruce, fir and pine

-Temperate

- moderate climate and relatively high rainfall
- Dominated by Deciduous trees such as oak, beech and maple

-Tropical Rain Forest

- highest degree of biodiversity
- near the equator
- Philippine lowland forest: **Dipterocarp trees**

-Grassland

- dominated by grasses, US: Prairie
- Dry grassland and shrubs: Chaparral
- Mongolia: Steppe
- Africa: Savanna
- Chaparral (dry grassland and shrubs)

-Deserts

- arid conditions
- hot and cold

Evolution

Mimicry:

Batesian mimicry, a harmless mimic poses as harmful. e.g Monarch and viceroy butterfly

Müllerian mimicry, two or more harmful species mutually advertise themselves as harmful. e.g bumblebees

Geographical isolation

- with physical barriers leading to speciation
- allopatric speciation

Behavioral Isolation

- without physical barrier but still leads to speciation (behaviors like mating preferences)
- Sympatric speciation

Homologous structure- similar structure, different function (e.g human arm, bird's wing, dog's foot)

Analogous structure- similar function, different structure (e.g bat's wing, butterfly's wings)

Theory of Evolution

(Jean Baptiste de Lamarck)

-Theory of Need

- Production of new organism arises from the need of it

-Theory of Use and Disuse

- Continued use of an organ strengthens it and continued disuse weakens it until it disappears

-Theory of Inheritance of Acquired Characteristics

- Characteristics that have been acquired be change during the lifetime of the organisms is transmitted to the offspring

Theory of Natural Selection

- Charles Darwin
- Differential survival and reproduction of individuals due to differences in phenotype.
- "Survival of the Fittest" (Spencer)

Other Mechanisms:

- Mutation (change in genetic composition)
- Genetic drift (Random change in the frequency of genes/allele, causing reduction of genetic variation)
- Gene flow (transfer of genetic variation; immigration and emigration)

Genetics

- Science of heredity
- Gregor Mendel (father of genetics)
- Genes – basic unit of inheritance
- Genetic makeup: genotype
- Observable traits: phenotype
- DNA – molecular basis of genes
- Chromosomes- composed of DNA and proteins
- Genome – all DNA in an organism